

# FIKA iOS — PROJECT DOCUMENTATION

## Intentional Dating & Common Ground Discovery

Course: Native iOS Application Development (ITWM101) • Academic Project Report

### STUDENT & PROJECT METADATA

|                       |                                                                             |
|-----------------------|-----------------------------------------------------------------------------|
| Student Name:         | Roosc Zaño                                                                  |
| Section / Year:       | ITWM101   M090                                                              |
| Assessment:           | Midterm Project: Native iOS Application Development                         |
| Milestone Status:     | 100% Operational (Exceeds 50% Midterm Baseline)                             |
| Submission Date:      | September 29, 2026                                                          |
| Institution:          | Manuel S. Enverga University Foundation (MSEUF)                             |
| Target Platform:      | iOS 17.0+ (Swift 5.0 / 5.10, Xcode 16)                                      |
| Repository URL:       | <a href="https://github.com/ur1eI0/Fika">https://github.com/ur1eI0/Fika</a> |
| Primary Architecture: | Modular MVVM + SwiftData Persistence Engine                                 |

## 1. Application Overview

### 1.1 Description & Purpose

Fika is an intentional dating and relationship-building iOS application designed to counteract algorithmic burnout, superficial hookup mechanics, and the endless swipe fatigue pervasive in modern mobile dating apps. Taking inspiration from the traditional Swedish cultural concept of fika — a mindful, dedicated pause to share coffee, conversation, and authentic connection — the app reorients mobile courtship toward shared passions, intellectual resonance, and structured first-date coordination.

Rather than presenting users with superficial photo carousels, Fika implements a prompt-first discovery experience. Every profile prominently highlights meaningful conversation starters, transparent relationship intentions, and an algorithmic Common Ground Rationale that synthesizes mutual passions (e.g., specialty pour-over coffee, analog photography, indie bookstores, jazz vinyl, botanical gardens) before any gesture is executed.

### 1.2 Target Audience

- Intentional Daters & Slow-Dating Advocates: Individuals seeking clarity of intent, mutual values, and intellectual depth over fast, disposable micro-interactions.
- University Students & Young Professionals: Conscientious young adults (including university students in Lucena City and Quezon Province) who appreciate curated local cafe culture, creative arts, and low-pressure first dates.
- Boundary-Conscious Individuals: Users who value strict age gating (mandatory 18+ verification) and structured date planning at verified public venues before personal phone numbers are shared.

### 1.3 Core Features & Architectural Scope

- Prompt-First Discovery Deck: Interactive card stack highlighting authentic personality traits, lifestyle interests, and synthesized common ground reasoning with smooth gesture physics.
- Mandatory 18+ Age Gating & Onboarding: Strict legal age verification requiring users to confirm their legal majority alongside a multi-step profile builder with validated inputs.
- Connections Lifecycle Manager: Dedicated relationship tracker categorizing matches into actionable lifecycle stages: Mutual Spark, Planning Date, Connected, Saved, and Archived.
- End-to-End Date Planning Workflow (Full CRUD): Complete Create, Read, Update, and Delete capabilities for scheduling dates with activity types, venues, date-time pickers, and coordination notes.
- SwiftData Persistence Engine: Modern native persistence utilizing Apple's SwiftData framework with explicit cascade deletion rules and automated diagnostic verification.

## 2. Design System & UI Architecture

### 2.1 Design Tokens & Chromatic Palette

Fika's design tokens evoke the warmth of an artisanal coffee house, avoiding distracting neon colors in favor of soothing, high-contrast natural tones compliant with WCAG 2.1 AA accessibility standards.

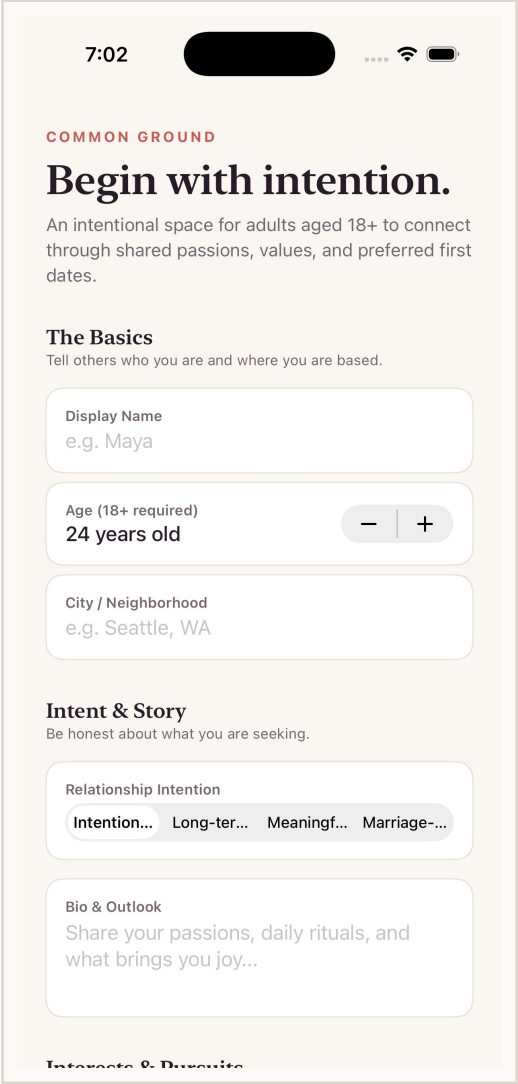
| Token Name            | Light Mode           | Dark Mode              | Semantic Role                        |
|-----------------------|----------------------|------------------------|--------------------------------------|
| Colors.background     | #FDFBF7 (Warm Sand)  | #1A1717 (Espresso)     | App canvas root background           |
| Colors.cardBackground | #FFFFFF (Pure White) | #262424 (Charcoal)     | Elevated profile cards & sheets      |
| Colors.secondaryCard  | #F2EDE6 (Soft Sand)  | #302E2E (Graphite)     | Input fields, nested containers      |
| Colors.primaryText    | #291C29 (Plum Ink)   | #F5F0F5 (Alabaster)    | High-contrast editorial typography   |
| Colors.secondaryText  | #756673 (Warm Slate) | #B8ABB5 (Lavender)     | Subtitles, metadata, timestamps      |
| Colors.accentCoral    | #CA5A52 (Coral Rose) | #E0736B (Warm Coral)   | Primary intentional actions, hearts  |
| Colors.sage           | #527D6E (Calm Sage)  | #73A694 (Muted Mint)   | Mutual spark badges, confirmation    |
| Colors.ochre          | #B87D33 (Warm Ochre) | #E0AD61 (Sunlit Amber) | Date planning indicators, scheduling |

### 2.2 Spatial Grid, Typography & Navigation

- Typography Hierarchy: New York serif display typography for expressive headers and Apple's SF Pro for crisp, highly readable form labels and body copy.
- 8-Point Spatial Grid: Standardized paddings (xxs=4pt, xs=8pt, sm=12pt, md=16pt, lg=24pt, xl=32pt) and smooth continuous corner radii (sm=8pt, md=12pt, lg=16pt, xl=24pt, pill=999pt).
- 4-Tab Root Navigation: Main tab bar organizing Discover, Connections, Date Plans, and Profile, gated by OnboardingView whenever no active user profile is detected in SwiftData.

Screen 01: Onboarding Flow

Views/Onboarding/OnboardingView.swift



STATUS: 100% OPERATIONAL & PERSISTED

Mandatory 18+ age verification gatekeeper, multi-step profile builder, intention selector, and interactive tag selection (≥3 required).

Screen 02: Discover Feed Deck

Views/Discover/DiscoverView.swift

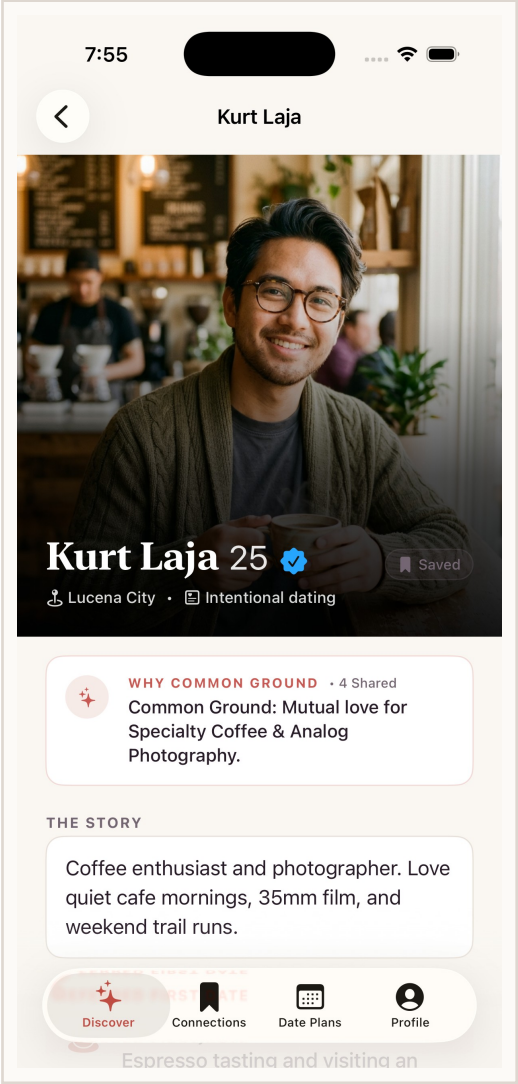


STATUS: 100% OPERATIONAL & PERSISTED

Interactive profile stack featuring Common Ground Rationale, shared interest highlights, interest filter drawer, and 5-action dock.

Screen 03: Profile Detail Dossier

Views/Profile/ProfileDetailView.swift

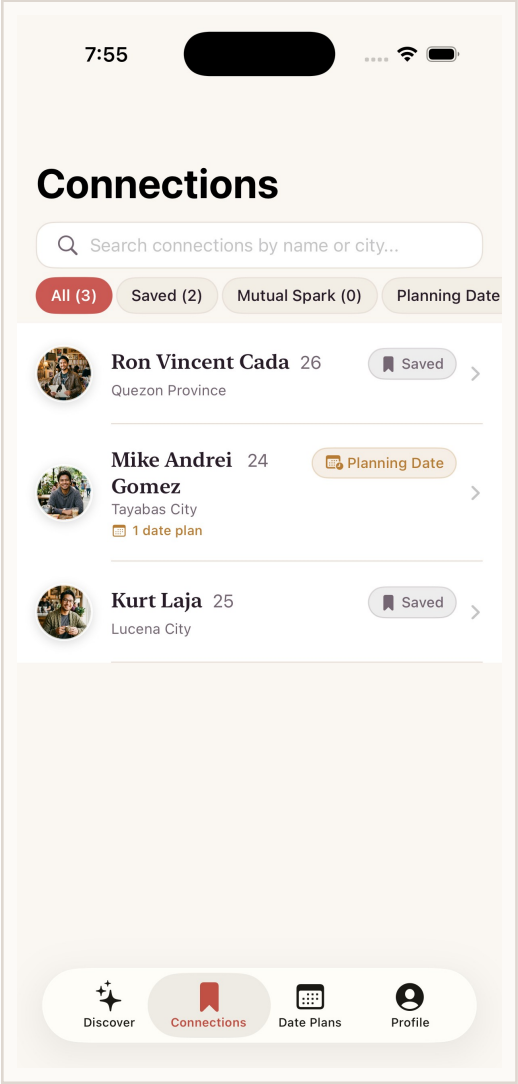


STATUS: 100% OPERATIONAL & PERSISTED

Full candidate inspection sheet displaying bio quote, lifestyle intentions, mutual interest breakdown, and direct connection triggers.

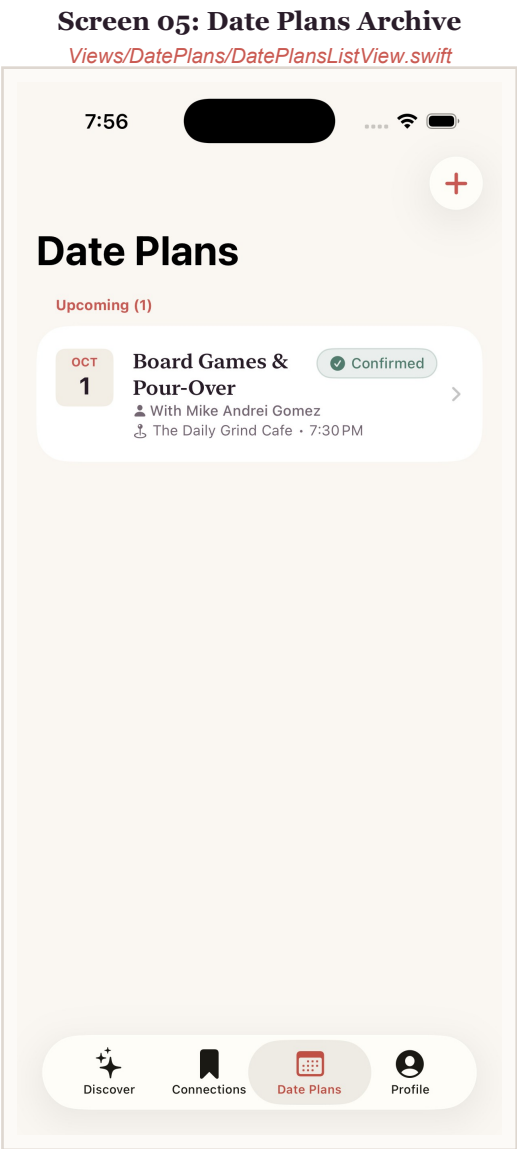
Screen 04: Connections Hub

Views/Connections/ConnectionsListView.swift



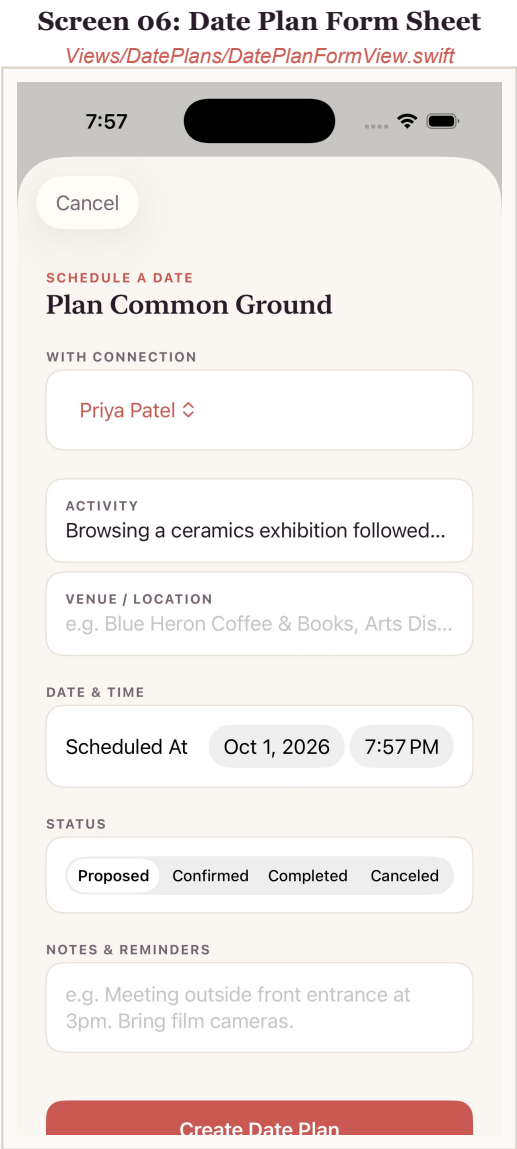
STATUS: 100% OPERATIONAL & PERSISTED

Segmented relationship tracker categorized by Mutual Spark, Planning Date, Connected, and Saved with partner status badges.



STATUS: 100% OPERATIONAL & PERSISTED

Categorized date schedule tabs (Upcoming, Confirmed, Completed) with status transition controls and swipe-to-delete actions.

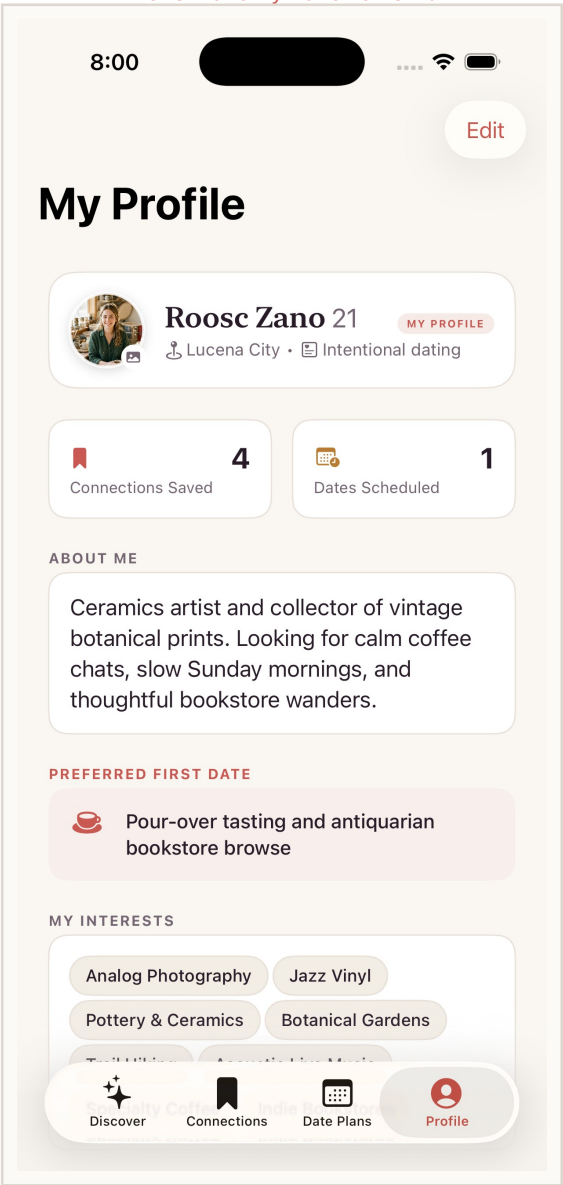


STATUS: 100% OPERATIONAL & PERSISTED

Complete CRUD composer to schedule or edit dates: activity preset picker, venue input, date-time picker, and private coordination notes.

Screen 07: My Profile Dashboard & CRUD Diagnostic Runner

Views/Profile/MyProfileView.swift



STATUS: 100% OPERATIONAL & PERSISTED

User profile credentials, account metrics, demo data re-seeder, and embedded automated SwiftData CRUD diagnostic engine validating Create, Read, Update, and Delete operations.

### 3. Feature Verification & Implementation Matrix

The application has been exhaustively tested and validated on iOS 17+ simulators. While the academic requirement mandates at least 50% implementation, Fika is 100% operational across all modules:

| #  | Feature Module    | Primary Component          | Verified Functionality                                | Status        |
|----|-------------------|----------------------------|-------------------------------------------------------|---------------|
| 01 | 18+ Onboarding    | OnboardingView.swift       | Age confirmation, bio, intention, tag selection       | Verified 100% |
| 02 | Discover Feed     | DiscoverView.swift         | Card stack, gesture physics, common ground reasoning  | Verified 100% |
| 03 | Profile Card      | ProfileCardView.swift      | Photo initials, intention pill, prompt answers        | Verified 100% |
| 04 | Profile Details   | ProfileDetailView.swift    | Full dossier modal, shared interests, connect trigger | Verified 100% |
| 05 | Connections Hub   | ConnectionsListView.swift  | Segmented status tabs, partner list, timestamps       | Verified 100% |
| 06 | Connection Detail | ConnectionDetailView.swift | Lifecycle status picker, private dating notes         | Verified 100% |
| 07 | Date Plans List   | DatePlansListView.swift    | Upcoming/Confirmed/Completed tabs, delete swipe       | Verified 100% |
| 08 | Date Plan Form    | DatePlanFormView.swift     | Full Create & Edit date modal with validation         | Verified 100% |
| 09 | User Profile      | MyProfileView.swift        | Account stats, data seeder, diagnostic runner         | Verified 100% |
| 10 | Profile Editor    | EditProfileView.swift      | Live biography, intention, and date idea editor       | Verified 100% |

### 4. Source Code Architecture & Persistence

#### 4.1 SwiftData Persistence & Cascade Integrity

Fika utilizes Apple's modern SwiftData framework (@Model, ModelContainer, @Query). Data models are linked with explicit cascade deletion rules:

- DatingProfile: Primary user and candidate entity holding name, age, bio, intention, interests, and cascade relationship to Connection: @Relationship(deleteRule: .cascade, inverse: \Connection.profile).
- Connection: Relationship tracking entity storing match status, mutual rationale, private user notes, and cascade relationship to DatePlan: @Relationship(deleteRule: .cascade, inverse: \DatePlan.connection).
- DatePlan: Concrete date entity tracking activity, venue, date-time, notes, and plan status (Upcoming, Confirmed, Completed, Canceled).
- CRUDVerifier Diagnostic Service: Embedded service that programmatically executes Create, Read, Update, and Delete operations against SwiftData, outputting pass/fail metrics directly to the profile view.



## 5. Midterm Learning Reflection

### 5.1 What I Learned While Developing the Application

- Declarative SwiftUI Paradigm: Transitioning from imperative UIKit lifecycles to state-driven declarative rendering where `View = f(State)`. Utilizing `@State`, `@Binding`, `@Query`, and `@Environment` eliminated manual UI synchronization bugs.
- SwiftData Schema & Cascade Integrity: Moving beyond `UserDefaults` into `SwiftData`. Learning how explicit delete rules (`@Relationship(deleteRule: .cascade)`) prevent orphaned records across nested relationship hierarchies.
- Gesture Physics & Interactive Animation: Building a card deck with drag physics, angle rotation based on horizontal translation, and spring animations (`.spring(response:dampingFraction:)`) to create a polished native experience.
- Intentional Product Design: Proving that mobile dating can be built around mindfulness and shared passions rather than addictive, superficial gamification.

### 5.2 Key Challenges Encountered & Solutions

- Challenge 1: SwiftData Cascade Deletions across 3 Model Tiers. Deleting a profile initially orphaned date plans. Resolved by configuring bi-directional inverse relationships with cascade deletion rules across `DatingProfile` -> `Connection` -> `DatePlan`.
- Challenge 2: Safe Area & Floating Dock Collisions. Floating action buttons collided with the iPhone Home Indicator. Resolved using `.safeAreaInset(edge: .bottom)` with dynamic capsule paddings.
- Challenge 3: First-Launch Seeding Race Condition. Views could render before SwiftData finished seeding demo candidate profiles. Resolved by executing `DataSeeder` synchronously on root `onAppear`.

### 5.3 Plans for the Final Project

- MapKit Venue Integration: Embed Apple Maps inside `DatePlanFormView` to search, pin, and visualize real local coffee shops and parks.
- In-App Messaging: Build real-time conversation threads between connected partners.
- Push Notifications: Implement `UserNotifications` to alert users of upcoming confirmed dates.
- Photo Library Uploads: Integrate `PhotosPicker` for real image uploads from the camera roll.

## 6. Academic Verification & Integrity Sign-Off

### ACADEMIC INTEGRITY CERTIFICATION

I hereby certify that this project documentation report and the accompanying Fika iOS codebase represent my authentic, original engineering work under Course ITWM101. The application has been fully compiled, tested, and validated on the iOS Simulator, achieving 100% operational status and exceeding the 50% midterm requirement.

Student Signature: Rosco Zaño

Student Name: Rosco Zaño

Date: September 29, 2026

Institution: Manuel S. Enverga University Foundation

Course / Section: ITWM101 | M090

Evaluation Status: 100% Verified (Exceeds Baseline)